

Chuanbo (Pilgrim) Peng

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EDUCATION

BSc (Hons) in Statistics | 2+2 Programme | WES GPA: 3.89/4.0

Expected in 07/2026

The University of Nottingham, Nottingham, UK

09/2022-06/2026

Current Average Mark: 82/100

Key Modules: Optimization (88), Probability and Statistics 2 (86), Probability 3 (84), Statistical Inference (78), Differential Equations 1 (84), Vector Calculus and Electromagnetism (92), Complex Analysis (81), Real Analysis (85); Ongoing: Time Series Analysis, Multivariate Analysis, Scientific Computation and Numerical Analysis

RESEARCH EXPERIENCE

Offline Evaluation of Sequential Decision Algorithms

07/2025-09/2025

Supervisor: Prof. Osman Yagan, Carnegie Mellon University (Remote)

- ♦ Investigated whether Inverse Propensity Scoring (IPS) can serve as a reliable substitute for cumulative regret when evaluating bandit algorithms under finite-data settings.
- ♦ Designed controlled empirical studies comparing Explore-Then-Commit, UCB, and Thompson Sampling across varying horizon lengths and data availability conditions using real-world interaction datasets.
- ♦ Examined ranking consistency between IPS-based estimates and cumulative regret-based benchmarks.
- ♦ Diagnosed estimator instability through Effective Sample Size (ESS) analysis, separating variance-induced ranking reversals from intrinsic metric disagreement.
- ♦ Identified frequent ranking instability driven by overlapping confidence intervals even under moderate ESS levels, highlighting structural limitations of IPS as a decision metric in offline policy selection.

MATHEMATICAL PROJECT

Numerical Modelling and Optimisation of a 2D Thermostat-Heater Control System

Ongoing

Applied mathematical and modelling project (Constrained PDE optimisation), University of Nottingham

- ♦ Formulated a two-dimensional heat diffusion model with Newton cooling boundary conditions to study spatial temperature dynamics under localised heating.
- ♦ Designed and implemented multiple control strategies (bang-bang and bounded hysteresis) and analysed their stability and convergence behaviour under finite time horizons.
- ♦ Defined a global objective functional measuring deviation from target temperature and conducted coarse-to-fine grid search to optimise thermostat placement.
- ♦ Investigated optimisation landscapes under varying geometric configurations and boundary exchange coefficients, identifying structural shifts in minimiser location.

SKILL

Statistical Methods: Linear & Logistic Regression, Time Series Analysis, LSTM, Monte Carlo Simulation, Bayesian Methods

Optimisation: Convex optimisation, gradient-based methods

Programming: Python (NumPy, SciPy, Pandas, Matplotlib), LaTeX